

Year One Computer Eng, Computer Sc, Information Technology, and Information Systems

1. Represent the region of integration and evaluate

[4 marks]

$$\int_0^1 \int_{2y}^4 x^2 + y^2 dx dy$$

2. Show that the following vector field is conservative and find a suitable potential function ϕ . [4 marks]

$$\vec{F} = (2x + 1)y\vec{i} + (x^2 + x + 1)\vec{j}$$

3. Evaluate the following line integral

(a) using Green's theorem, (b) without using the Green's theorem.

[6 marks]

$$\oint_C (x^2\vec{i} + 4xy\vec{j}) \cdot d\vec{s}$$

where C is a closed path in the form of a triangle with vertices A(0,0), B(3,0) and (3,5).

4. (a) Determine the independent and dependent variable, order and degree of

[2 marks]

$$\frac{d^2 X}{dt^2} + 2X^3 = \cos t$$

(b) Solve the following Initial value Problem

[2 marks]

$$\frac{dy}{dx} = y^2, y(0) = 1$$

5. Solve the following Initial value Problem

[4 marks]

$$\frac{dy}{dx} + 2xy = e^{-x^2}, y(0) = -2$$

6. Find the general solution of

[4 marks]

$$\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} + 3y = 3e^{2x}$$

7. A body at an unknown temperature is placed in a room which is held at a constant temperature of $30^\circ F$. If after 10 minutes the temperature of the body is $0^\circ F$ and after 20 minutes the temperature of the body is $15^\circ F$. Find the unknown initial temperature. [4 marks]